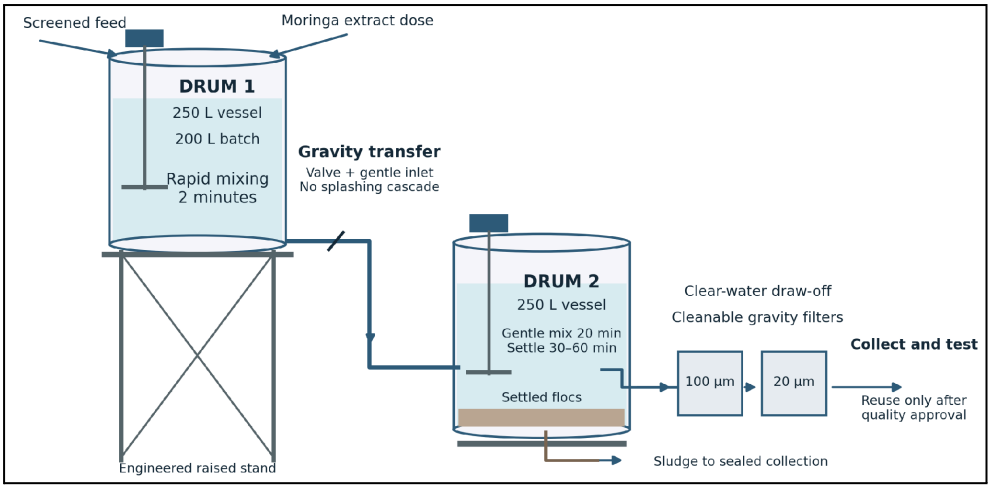
**Microplastice Removal using Moringa Protein**

**Community Filter Installation Details**

The system uses Moringa seed extract to bring suspended particles together, settle them and filter the clarified water. Two 250 L drums operate in sequence, treating a 200 L batch. Feed water is from the Ganges river .

**Why Moringa is used in the drum system**

The study tested Moringa seed extract and tamarind polysaccharide individually, not as a mixture.




The bioinformatics research and lab results established that with moringa microplastic particle-count reductions were 84.64% and 82.92% for tamarind at 100 mg/L in 50 mL polystyrene suspensions. Moringa is selected for the drum method because it gave the lower remaining count.

**Figure 1:** Moringa treatment with controlled gravity transfer, settling, filtration and separate sludge removal. Not to scale; operating settings and filter grades require verification.

**How the two drums work**

Drum 1 rapidly mixes the Moringa extract into the feed. Moringa proteins can help destabilise suspended particles through coagulation. Gentle mixing in Drum 2 brings particles together into larger clumps called flocs. The mixer then stops so settleable flocs collect at the bottom. Water is drawn from below surface scum and above sludge, then filtered. The method captures microplastics; it does not destroy them.

**Calculated starting dose**

| Final batch | Dose basis | Prepared extract | Stock at 10 g/L | Feed water |
|---|---|---|---|---|
| 100 L trial | 100 mg/L | 10 g | 1 L | 99 L |
| 200 L batch | 100 mg/L | 20 g | 2 L | 198 L |

Calculation: extract mass = 100 mg/L × 200 L ÷ 1,000 = 20 g. Stock volume = 20 g ÷ 10 g/L = 2 L. Make 20 g extract up to 2 L stock, then add to 198 L feed. The 200 L final volume leaves 50 L of headspace in each 250 L drum. Two drums treat one batch in sequence; capacity is not 500 L per batch.

**Working Method and Final Applications**

**Moringa Extract Preparation and Dosing**

At a dose of 100 mg/L, a 200 L batch requires 20 g of Moringa extract, provided the dose is expressed as dry extract mass. The extract preparation and concentration must follow the same basis used in the laboratory study. Raw seed quantity depends on extraction yield and cannot be substituted directly for extract mass.

**One batch step by step**

| Step | Operation |
|---|---|
| 1 Prepare | Screen feed through a removable 1 mm mesh. Record water source, volume, pH and initial particle count. Confirm the dose on a small sample of the actual feed before treating a full batch. |
| 2 Mix in Drum 1 | At the 100 mg/L starting dose, combine 198 L feed and 2 L stock. Rapidly mix for 2 minutes using a guarded variable-speed paddle mixer. |
| 3 Transfer | Immediately transfer to Drum 2 through a valved 32–40 mm gravity line and a baffled inlet. Aim initially for 5–10 minutes; measure actual flow. Avoid a splashing free-fall. |
| 4 Form and settle flocs | Gently mix in Drum 2 for 20 minutes after filling. Stop the mixer; leave undisturbed for 30–60 minutes, extending the rest if the flocs have not settled. |
| 5 Draw and filter | Slowly draw clarified water through 100 µm then 20 µm cleanable, low-head filters. Initial flow target: 3–5 L/min, subject to available gravity head and clogging. Stop before scum or sludge enters. |
| 6 Clean and record | Retain roughly 20 L as an initial bottom-water allowance; collect sludge and filter washings in a sealed container for approved disposal. Clean between batches; do not return captured solids to the drain. |

The study used 2 minutes rapid mixing, 20 minutes gentle mixing and 30 minutes settling. Its orbital-shaker speeds cannot be copied directly to drum paddles. Set mixer speed and dose through small-sample and drum tests. The split between drums and longer settling allowance require verification

**Hardware and realistic capacity**

Use two lidded HDPE drums, guarded paddle mixers, valves, sampling taps, sludge drains and accessible filters. Engineer the stand for full operating loads. Initially allow 0.5 m head between Drum 1's lowest operating level and Drum 2's maximum level; verify pipe and filter losses separately. Gravity moves water; mixers require power.

**Planning yield:** 200 L − 20 L retained = up to 180 L before other losses. The retained volume is an allowance, not measured sludge production. Filtering 180 L at 3–5 L/min takes 36–60 minutes. Allow roughly 2–3 hours per batch plus filling and cleaning; verify actual output.

**Leftover sludge and filter waste**

Sludge contains captured microplastics, Moringa and feed-water solids. Collect it in labelled, leakproof containers; capture filter waste and washings. Dewater only in a contained filter bag, collecting drainage for re-treatment. Use a local-authority-approved waste/sludge service and confirm the receiving facility accepts the tested material.

KUDABAZAR

**SWACHCHANJALI KUDA BAZAR FOUNDATION**

Non-Profit Organization

Reg. U88900UP2024NPL201283

# Letter of Recommendation

**Ref. - KBI/26/1** 1st October, 2026

To Whom It May Concern,

I am pleased to recommend Aarav Jain, who worked with Kuda Bazar to connect our Ganga clean-up initiatives with the development and pilot deployment of a community water-treatment system to remove microplastics from water. In my capacity as the director of Kuda Bazar, I had the opportunity to observe his efforts to translate an environmental research question into a practical engineering project. What distinguished Aarav was his willingness to take responsibility for the different stages of this work: understanding the treatment approach, building the mechanism to deliver it, and bringing the resulting prototype into a community setting.

When Aarav first explained his interest in water treatment, he traced it to his experience as a swimmer, describing how prolonged exposure to chlorinated pool water had left his hair brittle and his skin dry. His earlier work addressing arsenic, heavy-metal contamination, and turbidity in community water supplies had deepened his commitment to improving water quality and inspired him to turn his attention to microplastics. Building on these experiences, he began exploring whether naturally derived materials could play a practical role in water treatment. This inquiry led him to investigate moringa as a potentially affordable, biodegradable material for removing microplastics from water.

He pursued this interest through his research, where his analysis identified promising interactions between moringa's coagulant protein and microplastics. This research provided the conceptual foundation for his next question: how could a small treatment system deliver these plant-based preparations consistently enough to support practical use outside a laboratory?

That question connected closely with our work at Kuda Bazar. We have organised weekend clean-up drives since 2019, mobilising thousands of volunteers and collecting more than 10,000 kilograms of plastic waste through drives along the Ganga and at Banaras Hindu University. Aarav recognised the value of this sustained effort while identifying a complementary challenge: collecting visible waste from the riverbanks does not, by itself, address the smaller plastic particles already present in water. He wanted his project to extend the purpose of our clean-up work by exploring a treatment approach directed at this less visible form of pollution.

To translate that idea into a device, Aarav brought together mechanical fabrication, electronics, and automated process control. His system incorporated two water tanks equipped with level sensors, alongside manually replenished reservoirs for the plant-based treatment preparations. A programmable logic controller coordinated the dosing. The intention was to make the treatment sequence more repeatable and reduce its dependence on an operator manually measuring and stirring every batch.

This development work culminated in the pilot deployment of the community filtration unit at Assi Ghat, on the Ganga near Varanasi, in July 2026, in collaboration with Kuda Bazar. The unit was designed to introduce the moringa-based preparations into collected water, mix them under controlled conditions, and allow particle aggregation and settling before subsequent filtration. By processing 3, 250-litre batches daily, the pilot had a designed treatment capacity of approximately 500–750 litres per day—sufficient to provide treated water to nearly 200 visitors daily. Its deployment created a practical setting in which the operation of the device and the suitability of the treatment approach could be assessed, while providing microplastics-free water for thousands of passersby. For our organisation, it brought a research-led dimension to our river-conservation work without losing the community focus that makes that work meaningful.

What I value most about Aarav's contribution is the continuity between his curiosity and his actions. A personal observation led him to investigate a scientific question; that investigation led him to build a device; and the device became the basis for a community pilot. He demonstrated an ability to move between disciplines while remaining focused on the purpose of the work. His contribution was not simply an idea for cleaner water, but the effort to make that idea tangible enough to examine, operate, and improve.

I recommend Aarav with confidence. He brings intellectual curiosity, resourcefulness, and a strong sense of responsibility to environmental problem-solving. I believe he will contribute meaningfully to a university community that values students who connect scientific inquiry with practical service.

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**Gaurav Mishra**
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